

Case Study on the Ductile-to-Brittle Transition

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ABSTRACT

The **ductile-to-brittle transition (DBT)** is a critical phenomenon affecting material performance under varying temperature and loading conditions.

This study investigates the DBT behavior of **A508 low-alloy ferritic steel**, which exhibits a pronounced transition due to its BCC crystal structure.

Experimental data in the temperature range of **-200°C to 100°C** is analyzed to understand the variation in fracture energy.

The influence of loading velocity is studied using three regimes:

- Quasi-static (1 $\mu\text{m/s}$)

- Intermediate (0.5 mm/s)

- Dynamic (5 m/s)

A **sigmoidal (logistic) regression model** is used to represent the transition behavior and estimate the DBTT.

A Strict Lower Bound Envelope approach is developed to ensure conservative and safe design limits.

The results indicate that increasing loading rate shifts the transition and requires robust safety-based modeling.

OVERVIEW

- Historical Context
- Introduction
- Physical Meaning of Ductile and Brittle Fracture
- Mechanism Behind Ductile-to-Brittle Transition (DBT)
- Why BCC Materials Show High DBT
- Factors Affecting DBT
- Case Study & Computational Analysis
- Extracted Data
- Numerical Fitting Approach
- Comparison of Fracture Energy at Different Loading Rates
- Safety Bound (Lower Envelope) Construction
- Applications of Fitted and Safe Curves
- Result
- Conclusion
- References
- Thank You

HISTORICAL CONTEXT

Early understanding of ductile-to-brittle transition (DBT) came from major structural failures.

Titanic Disaster (1912):

Steel became brittle in cold conditions ($\sim -2^{\circ}\text{C}$).

Instead of deforming, hull plates fractured suddenly.

High sulfur content increased DBTT, leading to catastrophic failure.



Liberty Ships (1942–1945):

First large-scale use of welded structures.

Over 1000+ fracture incidents reported.

Several ships broke into two without warning.

Failures mainly occurred in cold conditions, below DBTT.

Use of low fracture toughness steel (rimmed steel) increased brittleness.

Key Causes Identified:

Low temperature $\rightarrow$ brittle behavior

Poor fracture toughness

Welded joints allowed crack propagation

These failures highlighted that strength alone is not sufficient—fracture toughness is critical.

Led to the development of fracture mechanics and Charpy impact testing.



INTRODUCTION

Ductile-to-Brittle Transition (DBT)

The ductile-to-brittle transition (DBT) is a phenomenon in which a material changes its fracture behavior from ductile to brittle with decreasing temperature. At higher temperatures, materials undergo plastic deformation and absorb significant energy before failure, whereas at lower temperatures, dislocation motion is restricted and crack propagation dominates, leading to sudden brittle fracture.

Transition from ductile → brittle behavior

High temperature → plastic deformation, high energy absorption

Low temperature → crack propagation, sudden fracture

Common in BCC metals

Ductile-to-Brittle Transition Temperature (DBTT)

The ductile-to-brittle transition temperature (DBTT) is the temperature at which a material transitions from ductile to brittle behavior. It represents the midpoint of the transition region and is typically determined using impact energy versus temperature curves obtained from Charpy testing.

DBTT = temperature of transition

Represents midpoint of transition region

Determined using Charpy impact test

Above DBTT → ductile | Below DBTT → brittle

Depends on strain rate and material properties

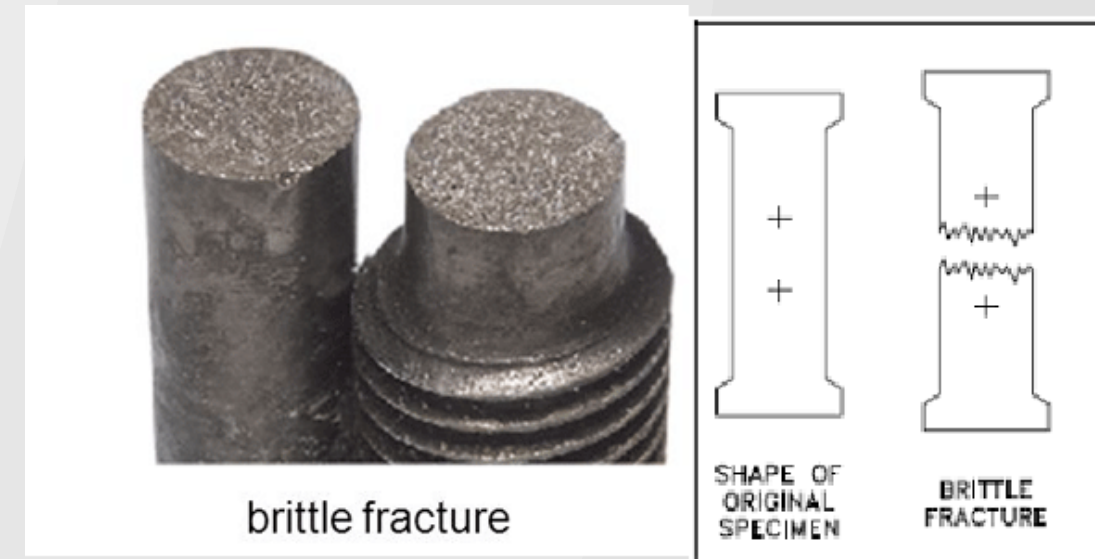
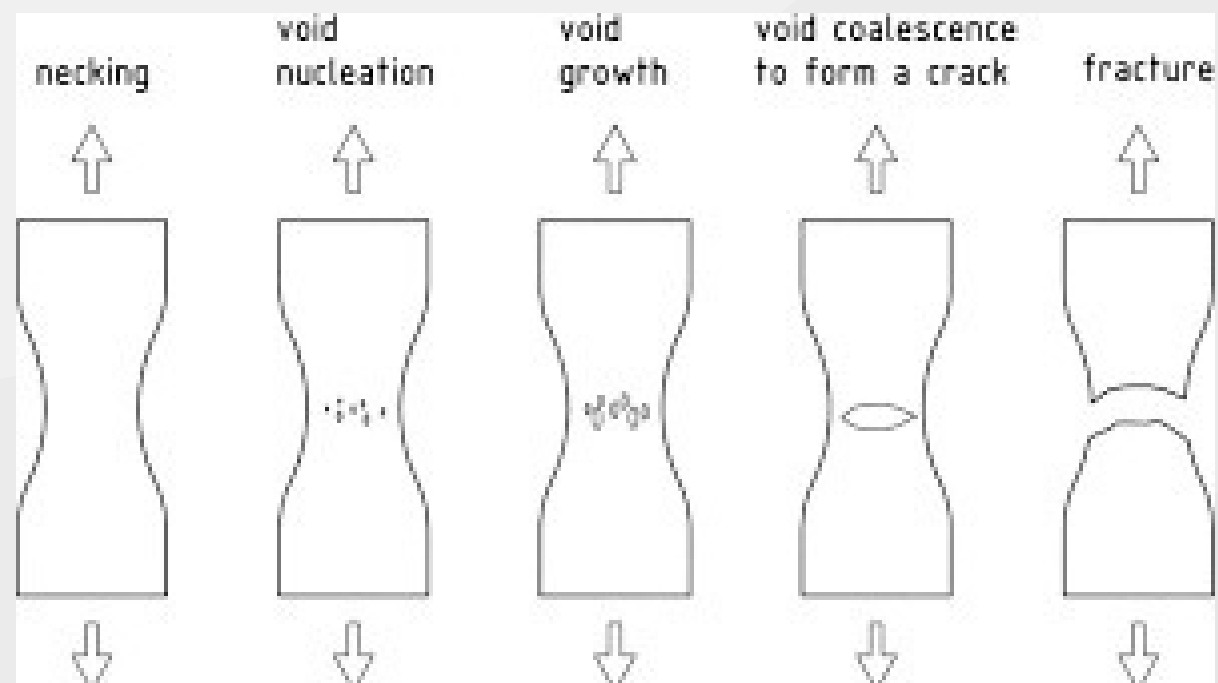
PHYSICAL MEANING OF DUCTILE AND BRITTLE FRACTURE

Ductile Fracture

Ductile fracture is characterized by significant plastic deformation before failure. The material first undergoes elastic and then plastic deformation, during which microvoids form, grow, and coalesce, leading to fracture. This process absorbs a large amount of energy and occurs in a slow and stable manner, providing visible warning before failure.

Key Points

- Large plastic deformation before fracture
- High energy absorption
- Microvoid formation and coalescence
- Slow and stable crack growth
- Failure is predictable



Brittle Fracture

Brittle fracture occurs with little or no plastic deformation, where the material fails suddenly under stress. In this case, crack initiation and propagation dominate, and fracture occurs by cleavage along specific planes. This type of failure involves very low energy absorption and happens rapidly without any prior warning.

Key Points

- No or very little plastic deformation
- Low energy absorption
- Crack propagates rapidly
- Cleavage-type fracture
- Failure is sudden and catastrophic

MECHANISM BEHIND DUCTILE-TO-BRITTLE TRANSITION (DBT)

The transition is governed by the competition between two physical mechanisms:

Dislocation Motion: Responsible for plastic deformation (Ductile behavior).

Crack Initiation & Propagation: Responsible for sudden fracture (Brittle behavior).

At Higher Temperatures (Ductile Regime)

Thermal Activation: High thermal energy facilitates the easy movement of dislocations within the crystal lattice.

Stress Relief: Stress is relieved through yielding and plastic deformation rather than crack growth.

Result: Crack growth is slow and stable, leading to energy-absorbing failure.

Dominance: Plastic deformation dominates over crack propagation.

At Lower Temperatures (Brittle Regime)

Restricted Mobility: Atomic vibrations reduce, making it harder for dislocations to move.

Increased Yield Strength: Higher stress is required for dislocation motion, causing yield strength to increase sharply.

Cleavage Fracture: Stress concentrates at crack tips because it cannot be relieved by yielding.

Result: Cracks propagate rapidly and unstably through atomic cleavage.

Dominance: Crack propagation dominates over plastic deformation.

Impact of Strain Rate (Loading Speed)

Time Dependency: Plastic deformation is a time-dependent process.

High Strain Rates: During impact loading, dislocations do not have sufficient time to move and accommodate deformation.

Shift in Dynamics: This raises the yield stress, causing the material to behave in a brittle manner even at temperatures where it might otherwise be ductile.

WHY BCC MATERIALS SHOW HIGH DBT

BCC (Body-Centered Cubic) materials, such as alpha-iron and mild steel, exhibit a pronounced ductile-to-brittle transition due to specific crystallographic characteristics.

1. High Peierls Stress

- > BCC materials possess a high Peierls stress, which represents a greater inherent resistance to dislocation motion.
- > This resistance makes plastic deformation difficult, especially as thermal energy decreases.

2. Strong Temperature Dependence

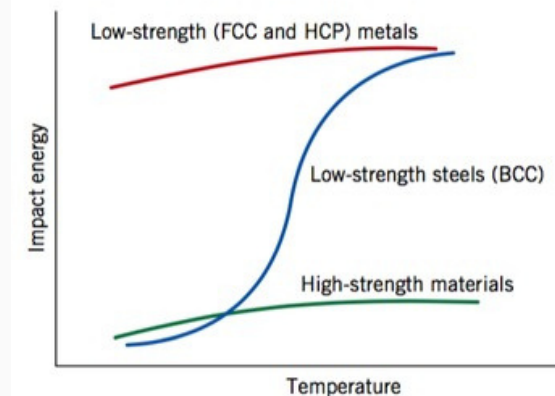
- > Dislocation motion in BCC structures is thermally activated.
- > At High Temperatures: Dislocations move easily, allowing for ductile behavior.
- > At Low Temperatures: Dislocation movement is severely restricted, leading to brittle behavior.

3. Absence of Close-Packed Slip Planes

- > Unlike FCC (Face-Centered Cubic) structures, BCC lacks close packed slip systems.
- > This results in fewer active slip systems at low temperatures.
- > Consequently, a much higher stress is required to initiate slip, which promotes brittle fracture instead.

4. Rapid Increase in Yield Stress

- > As the temperature decreases, the yield stress of BCC materials increases sharply.
- > In contrast, the fracture stress remains nearly constant regardless of temperature.
- > Transition Point: Brittle fracture occurs the moment the required yield stress exceeds the fracture stress.



Peierls Stress

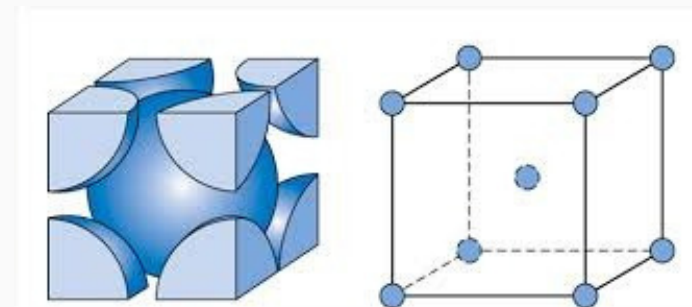
Peierls stress is the inherent internal resistance of a crystal lattice to the movement of a dislocation. It represents the energy barrier that a dislocation must overcome to move from one atomic position to the next within the slip plane.

BCC Material

A Body-Centered Cubic (BCC) material is a specific type of crystal structure where atoms are arranged in a repeating cube pattern. In this arrangement, there is one atom at each of the eight corners of the cube and one single atom positioned directly in the center of the cube.

Examples

- Mild Steel / Alpha-Iron
- Tungsten
- Chromium
- Molybdenum



FACTORS AFFECTING DBT

The transition temperature is not a fixed material constant; it is highly sensitive to the following metallurgical and environmental factors

1. Strain Rate (Loading Velocity)

Mechanism: Plastic deformation is time-dependent. Under high-velocity (impact) loading, dislocations do not have sufficient time to rearrange.

Effect: Increasing the strain rate shifts the DBTT to higher temperatures, making the material behave in a brittle manner even in warmer conditions.

2. Grain Size

Mechanism: Grain boundaries act as physical barriers to both dislocation movement and crack propagation.

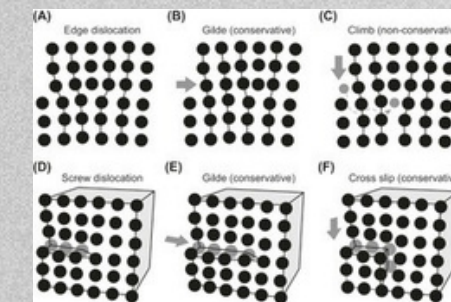
Effect: Finer grains (smaller grain size) improve toughness and lower the DBTT. This is one of the few ways to increase both the strength and the toughness of a material simultaneously.

3. Specimen Geometry & Notches

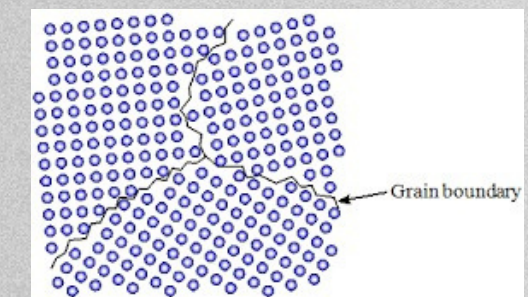
Mechanism: Sharp notches or cracks create a "triaxial stress state" and concentrate stress at the tip.

Effect: Notches inhibit plastic flow and promote brittle cleavage, effectively raising the transition temperature of the component.

Dislocations are tiny defects that act like "sliding joints" at the atomic level. They allow a metal to bend (ductility) by letting atoms slide past each other. If these joints "freeze" due to cold temperatures, the metal becomes brittle and breaks.



Grain boundaries are the interfaces where individual crystals (grains) of different orientations meet. They act as physical barriers that impede dislocation motion, which strengthens the material.



CASE STUDY & COMPUTATIONAL ANALYSIS

Description & Data Source:

The fundamental experimental basis for this study is derived from the research conducted by Benoît Tanguy et al.. The specific paper is titled "Ductile to brittle transition of an A508 steel characterized by Charpy impact test. Part I: experimental results".

Material Studied: A508 low-alloy ferritic steel, selected for its pronounced ductile-to-brittle transition characteristic of BCC crystal structures.

Experimental Method: The researchers utilized Charpy V-notch (CVN) impact tests to measure the energy absorbed during fracture across a wide temperature gradient from -200°C to 100°C.

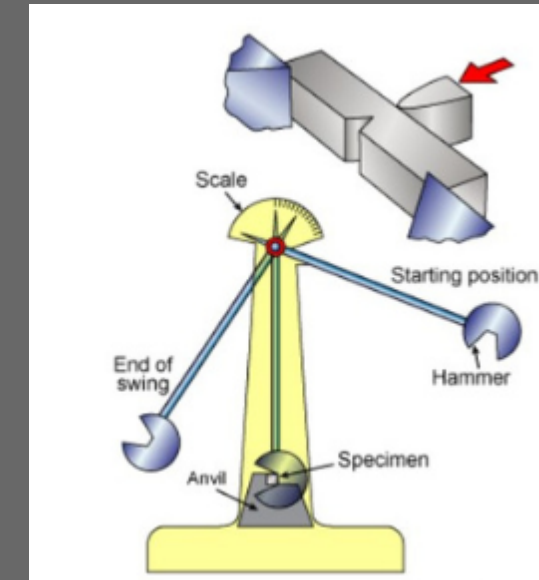
Loading Conditions: The study is unique in its use of three distinct loading velocities to evaluate strain rate sensitivity:

Quasi-static: 1 $\mu\text{m/s}$

Intermediate: 0.5 mm/s

Dynamic (Impact): 5 m/s

Data Representation: The dataset consists of digitized scatter points where multiple values at identical temperatures capture the stochastic variability and inherent material scatter within the transition zone.



- ✦ The Charpy V-Notch (CVN) test is a high-strain-rate experimental method used to measure a material's notch toughness and determine its transition temperature.
How it works:
 - >The Specimen: A 55mm long metal bar with a 2mm deep V-shaped notch is placed on a support.
 - >The Impact: A heavy pendulum is released from a height, striking the specimen directly behind the notch.
 - >The Measurement: The energy absorbed by the metal during fracture is calculated by how high the pendulum swings after the impact.

EXTRACTED DATA

Table shows the extracted Charpy impact energy data at different temperatures and loading velocity.

5 m/s		0.5 mm/s		1 μm/s	
Temperature (°C)	CVN Fracture Toughness (J/cm²)	Temperature (°C)	CVN Fracture Toughness (J/cm²)	Temperature (°C)	CVN Fracture Toughness (J/cm²)
-170.124	4.109589	-200	2.054795	24.06639	193.1507
-124.066	6.164384	-163.9	2.054795	-24.4813	195.2055
-126.556	4.109589	-143.983	4.109589	-43.1535	174.6575, 137.6712
-100.005	12.32877, 69.86301, 22.60274	-122.822	26.71233	-63.0705	94.52055, 43.15068
-91.7012	20.54795, 24.65753	-102.905	32.87671, 20.54795	-82.9876	39.0411, 73.9726
-92.9461	49.31507, 43.15068	-81.7427	53.42466	-102.905	45.20548
-81.7427	18.49315, 65.75342	-82.9876	30.82192	-124.066	32.87671
-82.9876	53.42466, 94.52055, 73.9726	-63.0705	53.42466, 102.7397	-143.983	26.71233
-63.0705	59.58904, 102.7397, 117.1233, 137.6712	-46.888	106.8493, 104.7945, 127.3973	-158.921	2.054795
-66.805	59.58904	-21.9917	115.0685, 147.9452	-201.245	2.054795
-61.8257	174.6575, 150	26.55602	211.6438		
-60.5809	129.4521	21.57676	213.6986		
-65.5602	154.1096				
-48.1328	189.0411, 176.7123, 170.5479, 147.9452, 90.41096				
-46.888	119.1781				
-30.7054	162.3288				
-31.9502	176.7123, 184.9315				
-20.7469	182.8767				
-19.5021	191.0959				
-21.9917	209.589				
-0.82988	217.8082, 250.6849				
-2.07469	271.2329				
21.57676	258.9041				
101.2448	269.1781, 256.8493				

NUMERICAL FITTING APPROACH

To transform scattered experimental data into a continuous analytical form, a **Logistic (Sigmoidal) Function** is employed. This model is ideal because it accurately represents the "S-shaped" behavior of the ductile-to-brittle transition as it shifts from a lower-shelf brittle region to an upper-shelf ductile region.

Mathematical Model: The energy-temperature relationship is defined as:

$$E(T) = \frac{L}{1 + e^{-k(T-T_0)}}$$

L (Upper Shelf Energy): The maximum fracture energy absorbed.

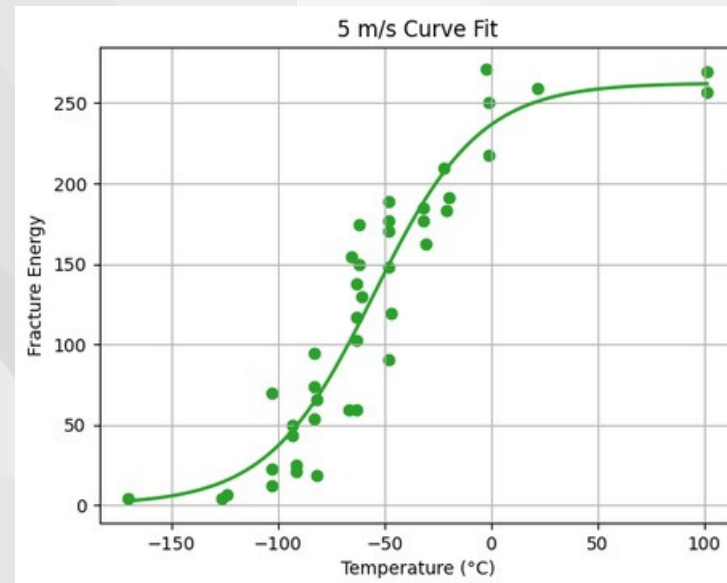
k (Slope Parameter): Controls the steepness/width of the transition zone.

T₀ (DBTT): The midpoint of the curve, identified as the Ductile-to-Brittle Transition Temperature.

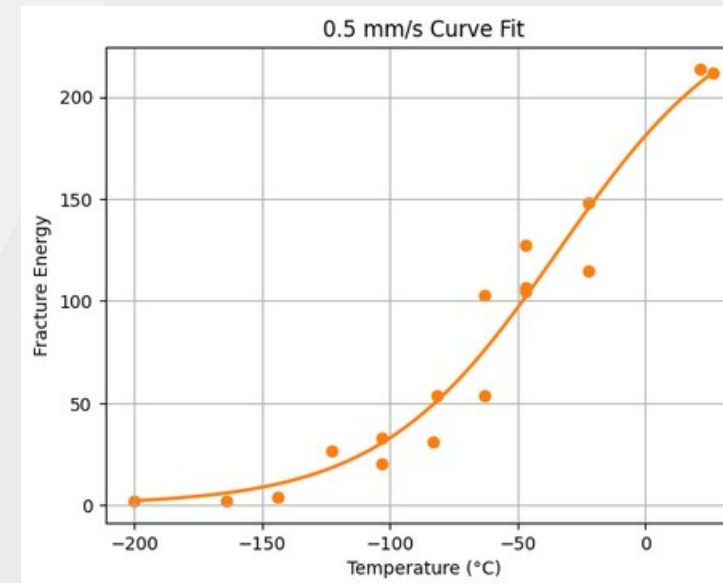
Computational Implementation: The fitting is performed using **Nonlinear Regression in Python** (SciPy library). By minimizing the residual difference between experimental scatter and model predictions, we generate a smooth, predictive curve that stabilizes the inherent variability of the raw data.

COMPARISON OF FRACTURE ENERGY AT DIFFERENT LOADING RATES

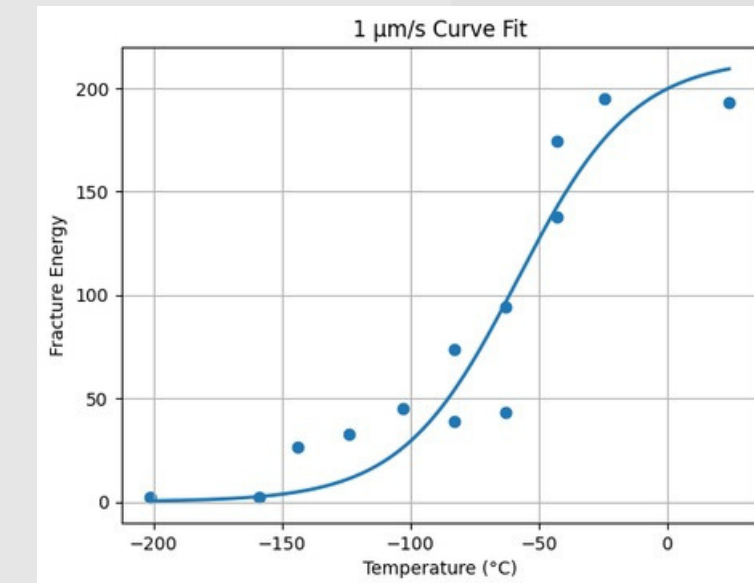
Dynamic Loading Graph (5 m/s)



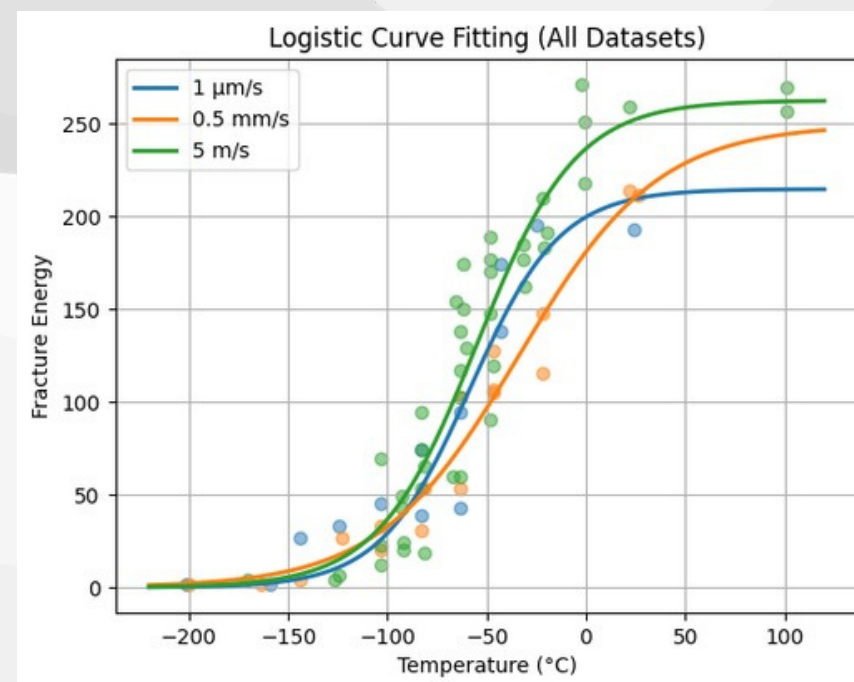
Intermediate Loading Graph (0.5 mm/s)



Quasi-static Loading Graph (1 μm/s)



Logistic Curve Fitting and Comparative Analysis (All Datasets)



As temperature rises, the energy required for fracture increases across all loading rates

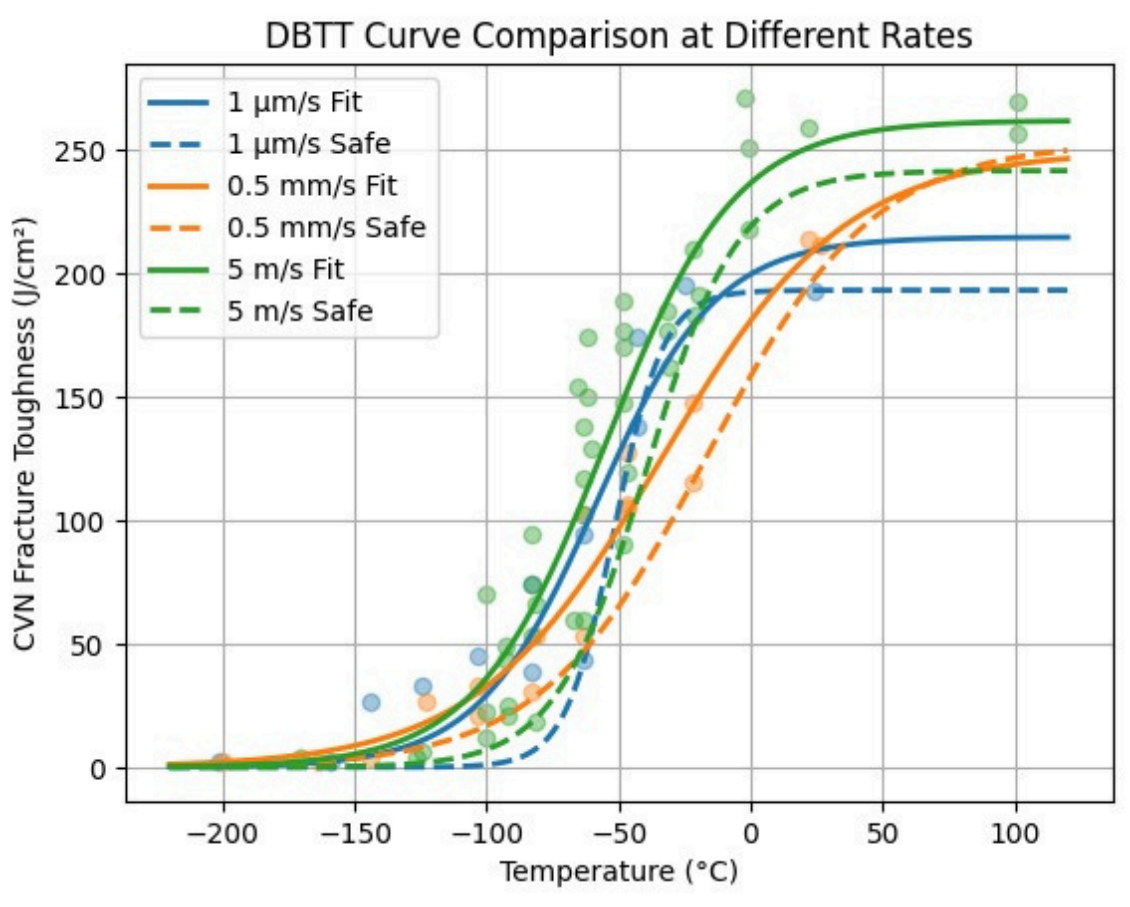
At Dynamic loading (5m/s), the metal becomes "stiffer" and harder to deform. This allows it to hold more load and absorb more energy before it finally snaps

The curve saturates at both ends because the material reaches its minimum and maximum possible toughness for that specific microstructure

SAFETY BOUND (LOWER ENVELOPE) CONSTRUCTION

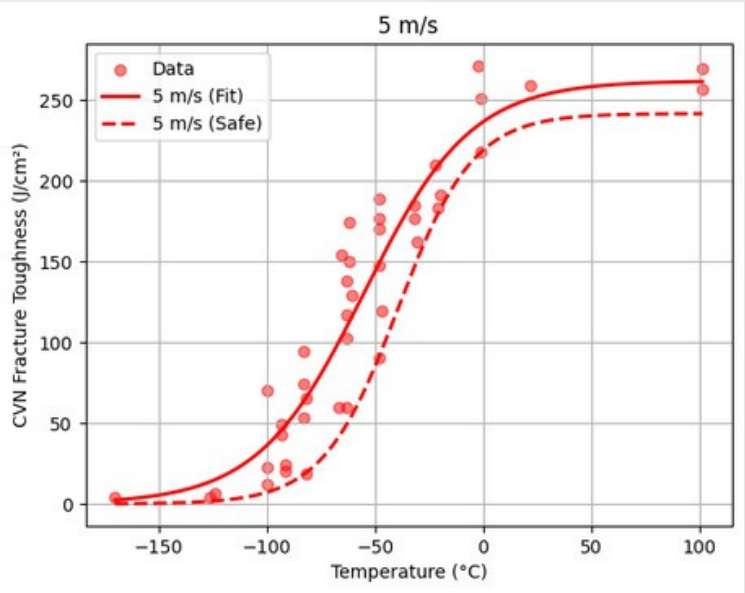
Due to the inherent scatter in Charpy impact energy data, particularly in the transition region, a conservative design approach is adopted by constructing a lower bound envelope. Instead of relying on average or fitted values alone, the minimum observed energy values at each temperature are considered.

Logistic Curve Fitting and Comparative Analysis (All Datasets)

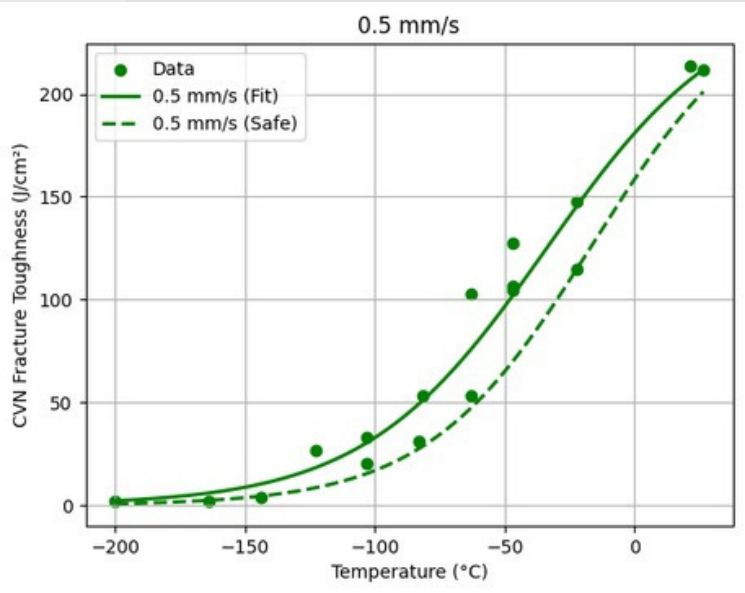


This lower bound is represented as a dashed curve below the scatter data and fitted sigmoidal trend. It is obtained by selecting the **lowest energy values** corresponding to each temperature and fitting them using a similar sigmoidal function. The resulting curve represents the minimum fracture resistance of the material under the given conditions.

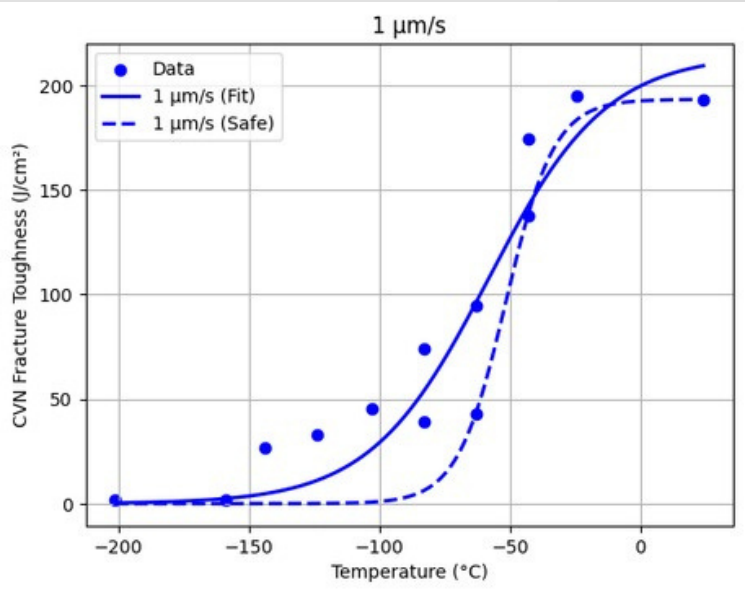
Dynamic Loading Graph (5 m/s)



Intermediate Loading Graph (0.5 mm/s)



Quasi-static Loading Graph (1 $\mu\text{m/s}$)



APPLICATIONS OF FITTED AND SAFE CURVES

The developed curves can be used to **predict material behavior and ensure safe design** under varying temperature and loading conditions.

Fitted Curve (Actual Behavior)

Represents average material response

Used for:

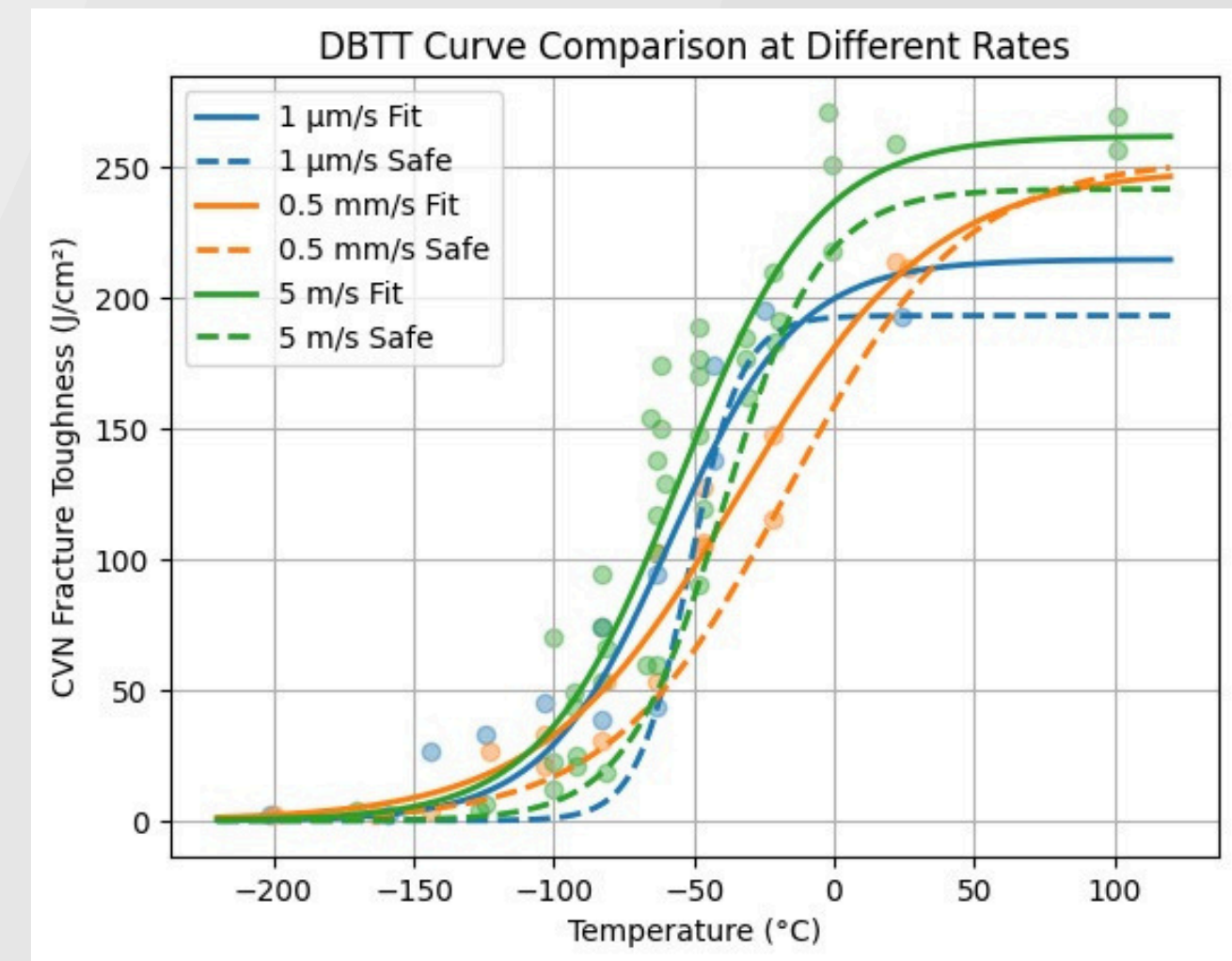
- Predicting fracture energy at given temperature
- Estimating DBTT
- Understanding material performance trends
- Simulation and modeling purposes

Safe Curve (Lower Bound / Conservative Design)

Represents minimum expected performance

Used for:

- Safety-critical structures (no failure allowed)
- Design of pressure vessels, pipelines, ships
- Ensuring operation below critical limits
- Engineering design with factor of safety



RESULT

The curves exhibit a sigmoidal variation of fracture toughness with temperature, clearly showing the ductile-to-brittle transition. At low temperatures, the material remains in the brittle region with low energy values. As temperature increases, a sharp rise in energy is observed in the transition region

The sigmoidal DBT curve shows saturation at both low and high temperatures due to the dominance of a single deformation mechanism in those regions.

Lower Shelf (Low Temperature):

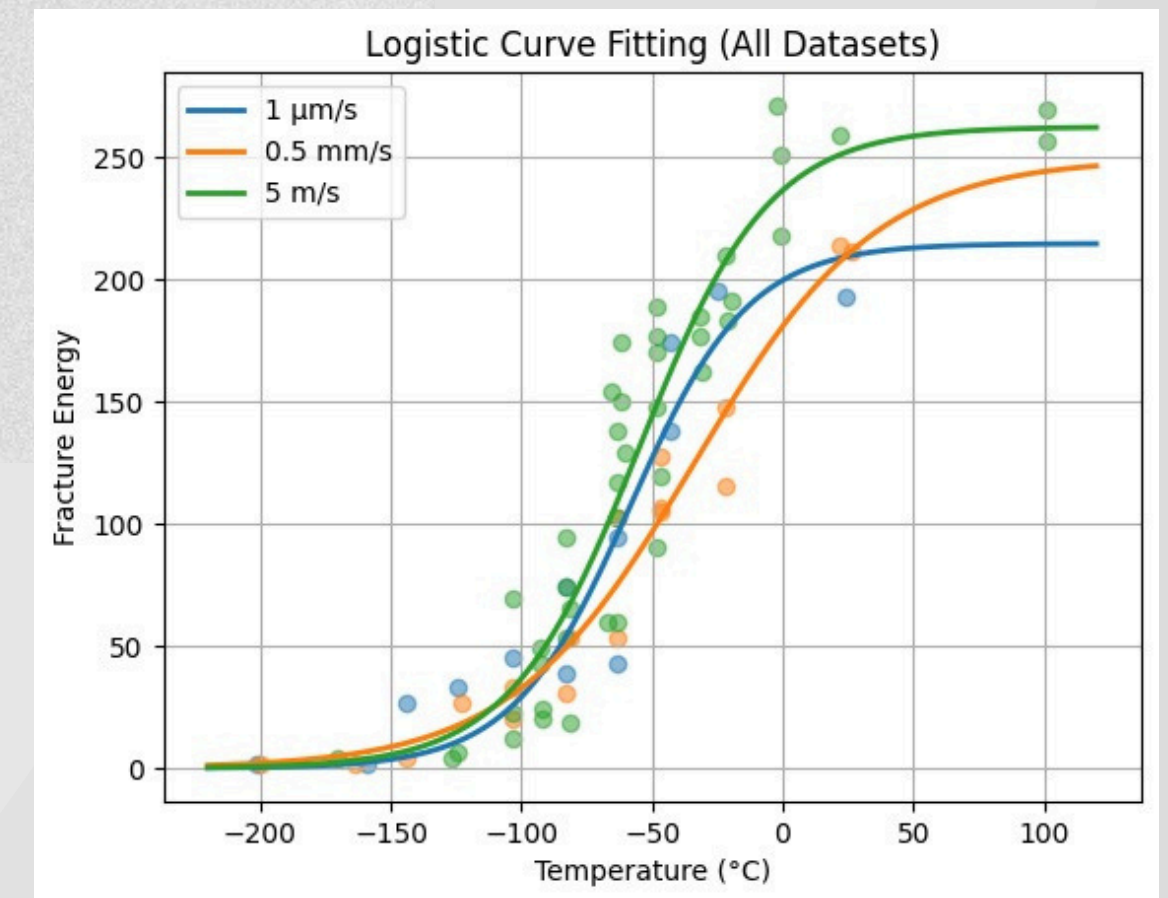
- Dislocation motion restricted
- Crack propagation dominates
- Energy remains constant → saturation

Upper Shelf (High Temperature):

- Easy dislocation movement
- Plastic deformation dominates
- Maximum energy reached → saturation

Curve crossing:

- Due to data scatter and stochastic (random / unpredictable) fracture behavior
- Common near DBTT where variability is high
- Does not affect overall trend or conclusions

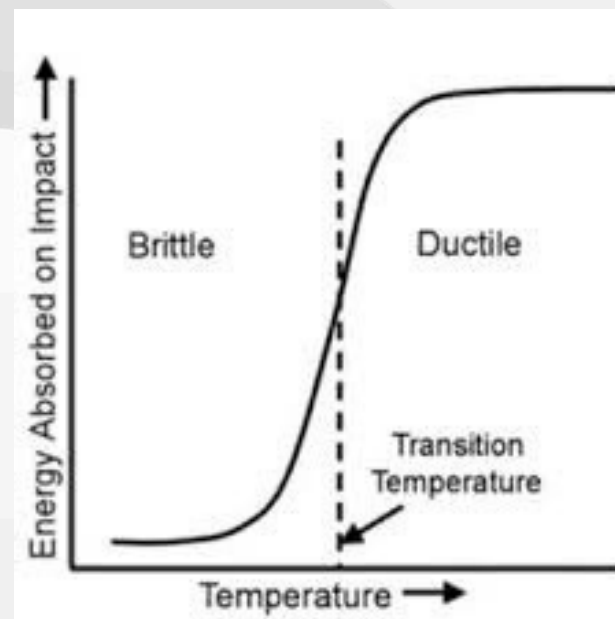
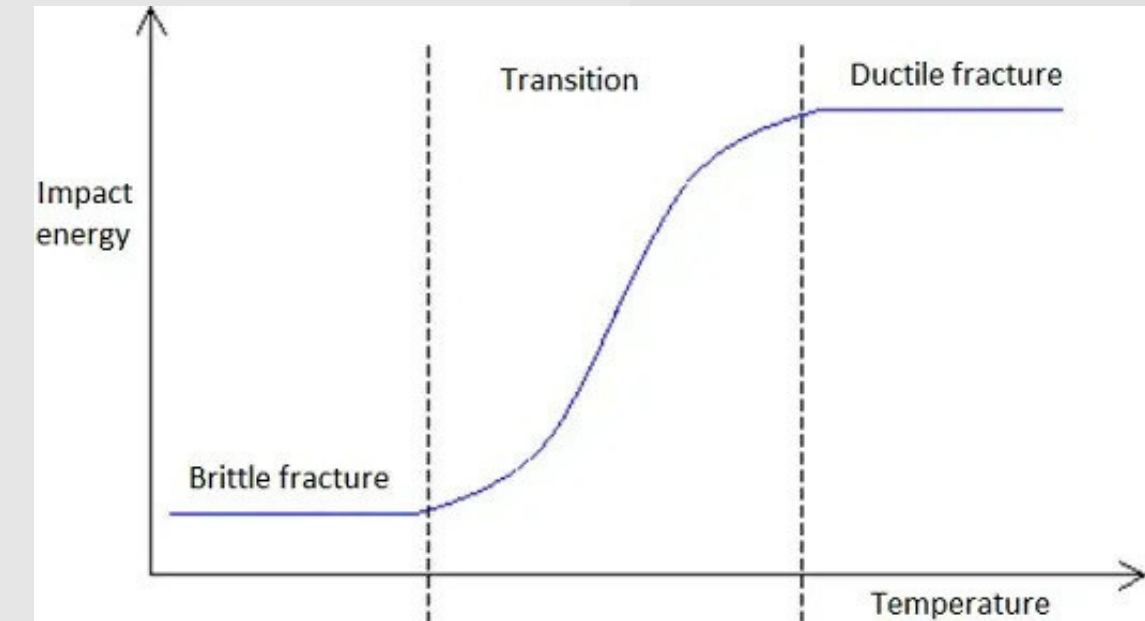


CONCLUSION

Transition Zone

The transition zone exists in the intermediate temperature range where the material behavior changes from brittle to ductile.

In this region, both crack propagation and plastic deformation occur simultaneously. Due to this competition, a **small change in temperature** or loading condition can cause a significant change in fracture energy, leading to a steep variation in the curve.



Transition temperature is the temperature at which the material changes its behavior from ductile to brittle.

It represents the midpoint of the transition zone, where both ductile and brittle fracture mechanisms are active.

Around this temperature, the material becomes highly sensitive, and small changes in conditions can significantly affect fracture behavior.

CONCLUSION

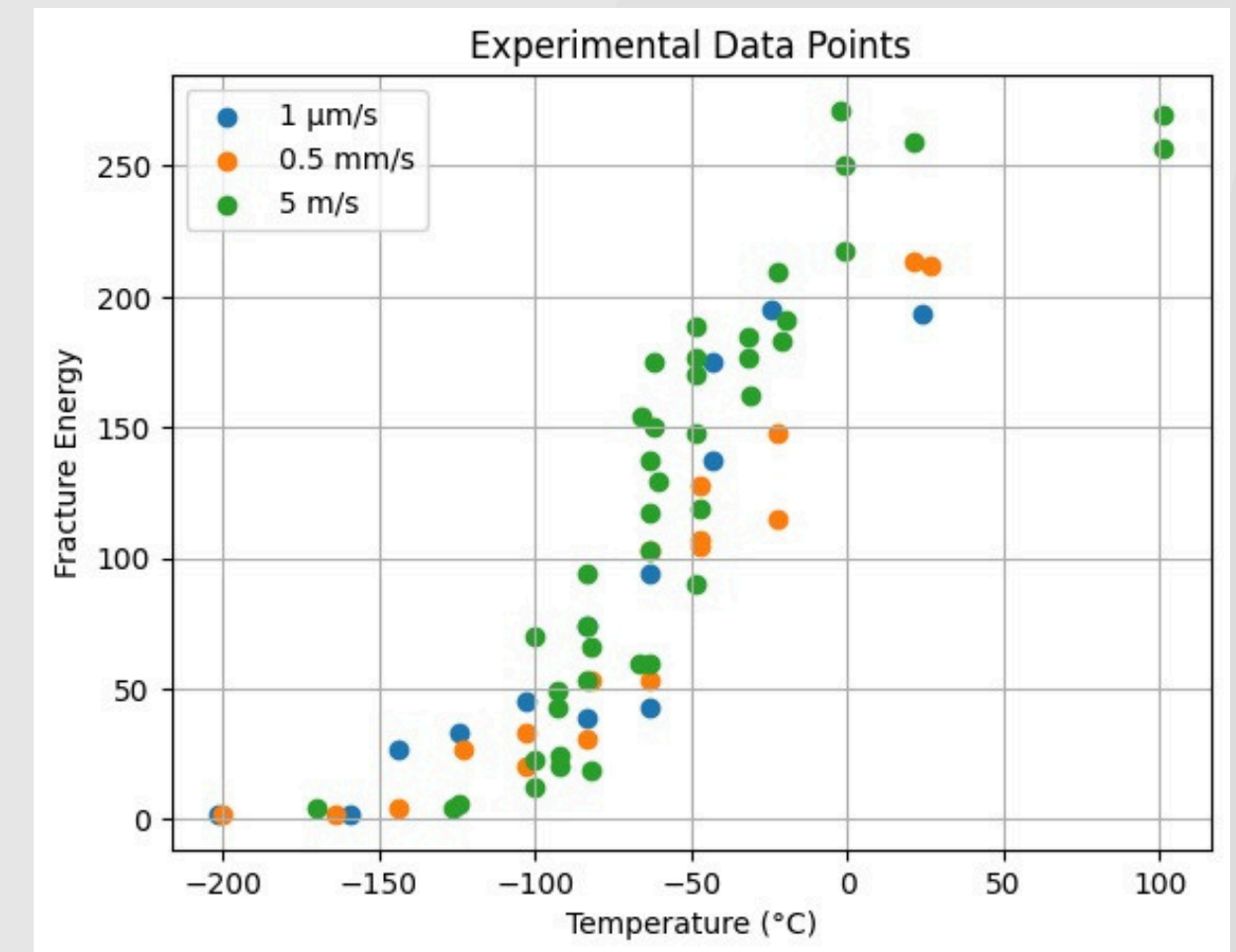
Why Data Scatter Occurs

Data scatter is more near the transition zone because fracture behavior becomes unpredictable. In this region, both ductile and brittle mechanisms compete, making the response highly sensitive to small variations.

- Microstructural variations (grain size, inclusions, defects)
- Different crack initiation sites and paths
- Competition between ductile and brittle mechanisms
- Sensitivity to strain rate and temperature
- Experimental uncertainties (specimen preparation, measurement errors)

How to Control / Handle Scatter

- Use statistical approaches instead of single-value analysis
- Apply curve fitting (logistic model) to capture overall trend
- Develop Lower Bound Envelope for safe design
- Increase number of experiments for better reliability
- Ensure consistent specimen preparation and testing conditions



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NUMERICAL INVESTIGATION OF THE TEMPERATURE DEPENDENCE OF DYNAMIC YIELD STRESS OF TYPICAL BCC METALS UNDER SHOCK LOADING WITH A DISLOCATION-BASED CONSTITUTIVE MODEL
AUTHOR LINKS
OPEN OVERLAY PANEL
SONGLIN YAO A, XIAOYANG PEI A, ZHANLI LIU B, JIDONG YU A, YUYING YU A, QIANG WU A
LOGISTIC CURVE FITTING AND PARAMETER ESTIMATION USING NONLINEAR NONITERATIVE LEAST-SQUARES REGRESSION ANALYSIS



THANK YOU
for your attention